

Quantum Project Docs Ready — Colab/Local Setup + Quick Test

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Here is a layout of the code and its usage:

Hi everyone,

I've put together documentation for the quantum project so you can run experiments on Colab, locally, or on GCP without getting stuck. I'm sorry this is coming a bit later than planned — I had multiple deadlines stacked this week — but everything should now be ready to go.

Documentation (start here)

All docs live in the repo with copy-pasteable examples:

- **README.md** — Overview, architecture, TL;DR, and quick start paths [README](#)
- **SETUP_COLAB.md** — Google Colab walkthrough (5 min, no local install needed) [SETUP COLAB](#)
- **SETUP_LOCAL.md** — Local development + GCP VM setup [SETUP LOCAL](#)
- **TROUBLESHOOTING.md** — Real bugs we hit + how to fix them [TROUBLESHOOTING](#)

Paths to Take:

- To run experiments → Colab (no install) [Setup Colab](#)
- Modify code → Local/GCP [Setup Local](#)
- Something's broken → [Troubleshooting](#)

Quick Test (About 5 minutes)

Option A: Google Colab (recommended, zero setup)

- Open the PROD notebook: [Quantum MAB Research Sandbox](#)
- In Colab, go to: **Runtime** → **Run all**
- Watch the progress bars; results auto-save to the shared Drive.
- Runtime: ~10–45 minutes depending on the config (the notebook has “quick test” presets).

Option B: Local terminal (≈15 minutes setup)

```
git clone https://github.com/pzg8794/quantum_project.git
cd quantum_project
python -m venv venv
source venv/bin/activate # or venv\Scripts\activate on Windows
pip install -r requirements.txt
# Run a quick test
bash scripts/run_exp_test.sh
```

The expected output is similar to Colab, but the results are saved under `./quantum_data_lake/` on your machine.

Key things to know

- **Shared data lake:** Results from Colab, local, and GCP all go to the same structure, so we can compare experiments easily.
- **Automatic checkpointing:** If Colab times out, reconnect and rerun — the framework resumes from the last checkpoint.
- **Cold-start warnings:** Messages like [WARN] No saved state... on first run are expected; that just means models are being trained from scratch.
- **Registry:** The first run is slower while the file index (registry) is built; subsequent runs are much faster.

Important links

- GitHub Repo: [quantum_project](#)
- PROD Notebook: [Quantum MAB Research Sandbox](#)

- Shared Data Lake: [Quantum_Data_Lake](#)
- Experiment Logs: [Quantum_Logs](#)
- Project Folder: [Drive_Quantum_Project](#)

If something's stuck

Check TROUBLESHOOTING.md first: [Github Troubleshooting Doc](#)

Run with verbose logging by adding verbose=True to your ExperimentConfiguration to see more detailed traces.

If you post in Slack, it helps to include:

- The error message/traceback
- Your config (models, scenarios, attack_type, runs, etc.)
- Whether you're on Colab, local, or GCP

Next steps

1. Pick Colab or local and run the quick test above.
2. Tweak the example configs in SETUP_COLAB.md or SETUP_LOCAL.md to try different models/scenarios.
3. Check the shared data lake to compare your runs with others'.
4. For larger batches, see the GCP section in SETUP_LOCAL.md.

TL;DR

Fastest: Open the PROD Colab notebook → Run all → results appear in the shared data lake.

Local: Clone → pip install -r requirements.txt → bash scripts/run_exp_test.sh.

Broken: See TROUBLESHOOTING.md or drop the error + config in Slack.

Let me know if you have any questions.

Best,

Piter